

Biology, 2018 EC, entrance exam, version 6

Grade 12 · Biology · 2018 E.C.

Time allowed: 120 minutes

Questions: 181

Total marks: 181

Name _____ Grade _____ ID No. _____ Date _____

1. What is the primary purpose of photosynthesis? (1)
A. To produce oxygen
B. To absorb sunlight
C. To produce glucose
D. To release carbon dioxide
-
2. Which organelle is responsible for photosynthesis? (1)
A. Mitochondria
B. Chloroplast
C. Nucleus
D. Ribosome
-
3. What is the chemical equation for photosynthesis? (1)
A. $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$
B. $6\text{O}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{CO}_2$
C. $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$
D. $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{CO}_2 \rightarrow 6\text{O}_2 + 6\text{H}_2\text{O}$
-
4. What is the primary purpose of cellular respiration? (1)
A. To produce glucose
B. To release energy from glucose
C. To absorb sunlight
D. To produce oxygen
-
5. Which gas is a byproduct of cellular respiration? (1)
A. Oxygen
B. Carbon dioxide
C. Nitrogen
D. Hydrogen
-
6. Where does cellular respiration occur in the cell? (1)
A. Chloroplast
B. Mitochondria
C. Nucleus
D. Ribosome
-
7. What is the chemical equation for cellular respiration? (1)
A. $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
B. $6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$
C. $6\text{O}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_5 + 6\text{CO}_2$
D. $6\text{H}_{12}\text{O}_6 + 6\text{CO}_2 \rightarrow 6\text{O}_2 + 6\text{H}_2\text{O}$
-
8. What is the role of chlorophyll in photosynthesis? (1)
A. To absorb sunlight
B. To release oxygen
C. To absorb carbon dioxide
D. To produce glucose
-
9. What is the role of ATP in cells? (1)
A. To store energy
B. To absorb sunlight
C. To produce oxygen
D. To release carbon dioxide
-
10. Which molecule is the final electron acceptor in the electron transport chain? (1)
A. Oxygen
B. Carbon dioxide
C. Glucose
D. Water

11. Which process occurs in the cytoplasm of cells? (1)

- A. Glycolysis** **B. Krebs cycle**
C. Electron transport chain **D. Calvin cycle**

12. Which molecule is the primary source of hydrogen for photosynthesis? (1)

- A. Glucose
B. Oxygen
C. Carbon dioxide
D. Water

13. Which process produces NADH and FADH₂? (1)

- A. Glycolysis
B. Krebs cycle
C. Electron transport chain
D. Fermentation

14. Which molecule is the primary source of electrons for cellular respiration? (1)

- A. Glucose B. Oxygen
C. Carbon dioxide D. Water

15. Which process produces the most carbon dioxide? (1)

- A. Glycolysis
B. Krebs cycle
C. Electron transport chain
D. Fermentation

16. Which process produces the most NADPH? (1)

- A.** Glycolysis
- B.** Krebs cycle
- C.** Electron transport chain
- D.** Light-dependent reactions

17. Which molecule is the primary source of energy for the Calvin cycle? (1)

- A. Glucose
B. ATP and NADPH
C. Carbon dioxide
D. Water

18. During oxidative phosphorylation, ATP is produced by combining ADP and inorganic phosphate using energy released from _____ (1)

- A. dehydrogenation of pyruvate to acetyl CoA.
B. transfer of electrons from NADH/FADH₂ to O₂
C. decarboxylation of citrate to oxaloacetate.
D. transport of electrons from complex IV to complex I.

19. Which one of the following comparisons between substrate level and oxidative phosphorylation is correct? (1)

During substrate-level phosphorylation

- A. 34 ATP molecules are produced, but in oxidative phosphorylation 4 ATP molecules are produced per glucose molecule.
- B. oxygen is the final electron acceptor, but in oxidative phosphorylation NAD is the electron acceptor.
- C. protons are pumped into the inter-membrane space, but in oxidative phosphorylation they are pumped into the cytosol.
- D. ATP is made in the cytosol and matrix, but in oxidative phosphorylation it is made on the inner membrane of mitochondria.

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- 20.** Which one among the following processes occurs during the light independent phase of photosynthesis? (1)
- A. Excitation of electrons from chlorophyll molecules
 - B. Reduction of triose molecules by hydrogen from NADPH
 - C. Splitting of water molecules into its components
 - D. Cyclic photophosphorylation of ADP in the thylakoid
-
- 21.** Which one of the following statements properly describes the structure of ATP? It is made up of (1)
- A. two adenine bases attached to a hexose sugar and three phosphate groups.
 - B. one adenine base attached to a ribose sugar and two phosphate groups.
 - C. two adenine bases attached to a deoxyribose sugar and two phosphate
 - D. one adenine base attached to a pentose sugar and three phosphate groups
-
- 22.** Which one of the following descriptions about glycolysis is correct? It is a process (1)
- A. that produces a net of 4 ATP through substrate level phosphorylation.
 - B. that takes place in the cytosol with high demand of oxygen
 - C. of breaking down of a glucose molecule to two pyruvates
 - D. of producing acetyl CoA from pyruvate by releasing CO₂
-
- 23.** What are the roles of grana and stroma in the process of photosynthesis? Grana is a site for (1)
- A. cyclic pathway while stroma is a site for non-cyclic pathway
 - B. photolysis of water while stroma is a site for carbon fixation
 - C. generation of sugar while stroma is a site for photolysis of water
 - D. production of NADPH while stroma is a site for chemiosmosis
-
- 24.** Which one of the following comparisons between cyclic and non-cyclic photophosphorylation is correct? Non-cyclic photophosphorylation (1)
- A. produces ATP, NADPH and O₂, but cyclic produces ATP and NADPH.
 - B. uses oxygen as the final electron acceptor, but cyclic uses cytochrome b-c1 complex.
 - C. uses photosystem I as electron donor, but cyclic uses water as an electron donor.
 - D. involves both photosystems I and II, but cyclic uses only photosystem I.
-
- 25.** All of the following events happen in the reduction stages of the Calvin cycle EXCEPT (1)
- A. combining of CO₂ with RuBP
 - B. utilization of ATP and NADPH molecules
 - C. conversion of 3-PGA into a sugar molecule
 - D. donating of electrons to a 3C- intermediates to form G3P
-
- 26.** What happens in the mitochondrion during the reactions of photorespiration? (1)
- A. Glycine is converted to serine with the release of CO₂ and NH₃.
 - B. Glycolate is converted to glyoxylate with the release of H₂O₂.
 - C. Glycerate is converted to GP with the release of ADP.
 - D. Phosphoglycolate is converted to glycolate with the release of Pi

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- 27.** How does photosynthesis ensure the continuity of life? By (1)
- A. constantly adding carbon dioxide to the atmosphere
 - B. maintaining carbon dioxide and oxygen balance in the atmosphere
 - C. constantly removing oxygen from the atmosphere
 - D. adding natural gas and absorbing methane from the atmosphere
-
- 28.** Which of the following is the difference between substrate level phosphorylation and oxidative phosphorylation? (1)
- Substrate level phosphorylation
- A. occurs in the inner mitochondrial membrane while oxidative phosphorylation occurs in the cytosol.
 - B. occurs during glycolysis and Krebs's cycle while oxidative phosphorylation occurs during electron transport chain.
 - C. has the same source of energy as oxidative phosphorylation.
 - D. occurs in aerobic respiration while oxidative phosphorylation in anaerobic respiration.
-
- 29.** Which one of the following is correct about the energy liberation phase of glycolysis pathway? It (1)
- A. hydrolyses two ATP molecules and raises the free energy of glucose.
 - B. breaks six carbon molecules into two glyceraldehyde-3-phosphate molecules.
 - C. releases one water molecule and dihydroxyacetone phosphate molecule.
 - D. produces four ATP, two NADH and two molecules of pyruvate.
-
- 30.** Which one of the following statements describes an event of the process of electron transport system correctly? (1)
- A. Oxygen serves as a terminal electron acceptor in the production of ATP.
 - B. Thirteen ATP molecules are invested when electrons are transferred from NADH/FADH₂ to O₂.
 - C. More than 90% of the total ATP is produced by substrate-level phosphorylation.
 - D. It happens due to the difference in pH as the inside mitochondrial membrane is more acidic
-
- 31.** In a complete oxidation of glucose molecule, how many net ATP molecules are produced by substrate level phosphorylation? (1)
- | | |
|------|-------|
| A. 2 | B. 4 |
| C. 8 | D. 10 |
-
- 32.** A carbohydrate compound is known to have 12 carbon atoms in the whole molecule. What could this compound be? (1)
- | | |
|---------------------|---------------------|
| A. A Polypeptide | B. A Disaccharide |
| C. A Monosaccharide | D. A Polysaccharide |
-
- 33.** Which of the following molecules is NOT a polymer? (1)
- | | |
|------------|-----------|
| A. Protein | B. Lipid |
| C. RNA | D. Starch |
-
- 34.** If certain food stuff gave a positive result with Benedict's test only after it was hydrolyzed with hydrochloric acid, which of the following substances could it be? (1)
- | | |
|-----------------------|-----------------|
| A. Non-reducing sugar | B. Nucleic acid |
| C. Lipid | D. Protein |
-
- 35.** At which level of structural organization do proteins have the alpha-helix shape? (1)
- | | |
|-----------------------|-------------------------|
| A. Primary structure | B. Secondary structure |
| C. Tertiary structure | D. Quaternary structure |

-
- 36.** What is the base found in RNA in place of thymine of DNA? (1)
- | | |
|-------------|------------|
| A. Cytosine | B. Guanine |
| C. Thymine | D. Uracil |
-
- 37.** Which element is found in nucleic acids? (1)
- | | |
|--------------|---------------|
| A. Calcium | B. Iron |
| C. Magnesium | D. Phosphorus |
-
- 38.** The most complex structure of proteins is called (1)
- | | |
|-----------------------|-------------------------|
| A. Primary structure | B. Secondary structure |
| C. Tertiary structure | D. Quaternary structure |
-
- 39.** In the formation of a macromolecule, what type of reaction would join two subunits together? (1)
- | | |
|--------------------------|-------------------------|
| A. Hydrolysis reaction | B. Dehydration reaction |
| C. Denaturation reaction | D. Hydrophobic reaction |
-
- 40.** Which are the four most abundant elements in living cells? (1)
- | | |
|---------------------------------------|---------------------------------------|
| A. carbon, oxygen, sulfur, phosphorus | B. carbon, oxygen, hydrogen, nitrogen |
| C. carbon, oxygen, nitrogen, sulfur | D. carbon, oxygen, sulfur, magnesium |
-
- 41.** Which of the following is an inorganic molecule? (1)
- | | |
|---|---|
| A. CaCO_3 | B. CH_4 |
| C. $\text{C}_6\text{H}_{12}\text{O}_{11}$ | D. $\text{C}_{18}\text{H}_{36}\text{O}_2$ |
-
- 42.** Which of the following is true about all proteins? (1)
- A. They are twenty amino acids long
 - B. They are globular in shape
 - C. They perform the same function
 - D. The monomers are held together by peptide
-
- 43.** Which of the following food types would most likely give a negative result upon addition of iodine solution? (1)
- | | |
|------------|-----------|
| A. Bread | B. Butter |
| C. Biscuit | D. Potato |
-
- 44.** Why does sucrose give a negative result when mixed and heated with Benedict's solution? (1)
- | | |
|--|--|
| A. Because it is not a sugar molecule. | B. Because it is a non-reducing sugar. |
| C. Because it is a reducing sugar. | D. Because it is not a monosaccharide. |
-
- 45.** Which one of the following is different from the other three? (1)
- | | |
|--------------|-------------|
| A. Starch | B. Glycogen |
| C. Cellulose | D. Maltose |
-
- 46.** Which level of protein structural organization involves two or more polypeptides? (1)
- | | |
|-----------------------|-------------------------|
| A. Primary structure | B. Secondary structure |
| C. Tertiary structure | D. Quaternary structure |

-
- 47.** Which of the following bases is NOT found in RNA? (1)
- | | |
|--------------------|-------------------|
| A. Adenine | B. Guanine |
| C. Cytosine | D. Thymine |
-
- 48.** When a drop of iodine solution is added to a solution of food stuff and the food stuff turns dark blue, what does the food stuff contain? (1)
- | | |
|--------------------------|------------------------|
| A. Reducing sugar | B. Disaccharide |
| C. Glucose | D. Starch |
-
- 49.** Among the following organic molecules, identify the one that does not contain nitrogen. (1)
- | | |
|------------------|-------------------|
| A. Starch | B. Protein |
| C. DNA | D. RNA |
-
- 50.** Which test is suitable for checking the presence of lipids in food stuffs? (1)
- | | |
|-------------------------|---------------------------|
| A. Biuret test | B. Benedict's test |
| C. Emulsion test | D. Iodine test |
-
- 51.** Which of the following is made of globular proteins? (1)
- | | |
|--------------------|--------------------|
| A. Enzyme | B. Keratin |
| C. Collagen | D. Glycogen |
-
- 52.** Which of the following units of measurement is more convenient to express the size of cellular organelles? (1)
- | | |
|----------------------|----------------------|
| A. Meter | B. Centimeter |
| C. Millimeter | D. Micrometer |
-
- 53.** Which of the following is NOT true about mitochondria and chloroplast? (1)
- | | |
|-------------------------------------|---------------------------------------|
| A. Both contain chlorophyll | B. Both have a double membrane |
| C. Both contain nucleic acid | D. Both transduce energy |
-
- 54.** Among the following scientists who contributed to the cell theory, identify the one who stated that 'a cell can arise only from another cell like it'. (1)
- | | |
|---------------------------|------------------------------|
| A. Robert Hooke | B. Matthias Schleiden |
| C. Theodor Schwann | D. Rudolf Virchow |
-
- 55.** What does it mean when biologist express the cell membrane as a unit membrane? (1)
- A.** A cell is covered by a single membrane
- B.** A membrane is only one lipid layer thick
- C.** All cells have essentially similar membrane
- D.** A membrane is covered by a single layer of protein
-
- 56.** Which of the following classes of molecules CANNOT pass easily across the cell membrane by simple diffusion? (1)
- | | |
|-------------------------------------|-------------------------------|
| A. Small non-polar molecules | B. Non-polar molecules |
| C. Lipid soluble substances | D. Polar molecules |
-
- 57.** Which of the following modes of transport is used by cells to move substances against their concentration gradients? (1)
- | | |
|---------------------------------|----------------------------|
| A. Osmosis | B. Simple diffusion |
| C. Facilitated diffusion | D. Active transport |

-
- 58.** Which of the following factors determines the rate at which organelles settle out of cell homogenate if spun in a centrifuge? (1)
- A. Mass of the organelle
 - B. Function of the organelle in the cell
 - C. Location of the organelle in the cell
 - D. Thickness of the membrane covering the organelle
-
- 59.** Which one of the following is the smallest of all? (1)
- A. A virus
 - B. An amoeba
 - C. A bacterium
 - D. A RBC
-
- 60.** What is the best term that expresses the movement of substance in cells against their concentration gradients? (1)
- A. Active transport
 - B. Passive transport
 - C. Osmosis
 - D. Diffusion
-
- 61.** Who was the person that first observed living cells moving around when they examined drops of water under the microscope? (1)
- A. Robert Brown
 - B. Anton Van Leeuwenhoek
 - C. Robert Hooke
 - D. Theodor Schwann
-
- 62.** In which of the following groups of living organisms do the cells lack organized nuclei? (1)
- A. Fungi
 - B. Protozoa
 - C. Bacteria
 - D. Algae
-
- 63.** Which of the following has a bigger size than all the others? (1)
- A. A ribosome taken from an animal cell
 - B. A mitochondrion taken from a plant cell
 - C. A nerve cell taken from a human brain
 - D. A glucose molecule taken from a plant cell
-
- 64.** Which of the following parts of the plant cell is NOT a living component of the cell? (1)
- A. Cell membrane
 - B. Cell wall
 - C. Cytoplasm
 - D. Nucleus
-
- 65.** Which of the following microorganisms are prokaryotic? (1)
- A. Bacteria
 - B. Protozoa
 - C. Yeast
 - D. Algae
-
- 66.** What happens when a plant cell absorbs in more water? (1)
- A. It shrinks
 - B. It becomes turgid
 - C. It becomes wilted
 - D. It swells and bursts
-
- 67.** Which of the following organelles is common to both plant and animal cells? (1)
- A. Cell wall
 - B. Chloroplast
 - C. Mitochondria
 - D. Central vacuole

-
- 68.** What happens when human red blood cells are kept in a hypertonic solution? (1)
- A. The same net gain and loss of water
 - B. More water gets into the cell than leaving it
 - C. More water goes out of the cell than getting in
 - D. The cells prevent water from getting in or leaving out
-
- 69.** Which knowledge about cell biology did scientists discover after all the others? (1)
- A. The first understanding that living things are made of cells
 - B. Knowledge about how chromosomes behave during mitosis
 - C. First understanding about the double helix structure of DNA
 - D. First understanding about the DNA sequence in human genome
-
- 70.** Which one of the following events happened before all the others? (1)
- A. The cell theory was proposed
 - B. The protozoa were discovered.
 - C. The compound microscope was invented.
 - D. The structure of DNA was described.
-
- 71.** When a plant cell is placed in a hypertonic solution, what will happen to the water within the cell and what is this process called? (1)
- A. leave the cells by osmosis, turgidity
 - B. move into the cells by osmosis, plasmolysis
 - C. move into the cells by osmosis, turgidity
 - D. leave the cells by osmosis, plasmolysis
-
- 72.** What mineral makes bones strong? (1)
- A. Oxygen
 - B. Calcium
 - C. Iron
 - D. Zinc
-
- 73.** What are the two bones in the lower arm? (1)
- A. Radius and ulna
 - B. Femur and humerus
 - C. Tibia and Fibula
 - D. Mandible
-
- 74.** The purpose of white blood cells is to (1)
- A. Carry oxygen
 - B. Speed up digestion
 - C. Protect against infection
 - D. All
-
- 75.** Which of the followings is a hinge joint? (1)
- A. Shoulder
 - B. Knee
 - C. Hip
 - D. None
-
- 76.** What is a fibrous connective tissue that attaches bone to bone? (1)
- A. Cartilage
 - B. Ligament
 - C. Tendon
 - D. Rib
-
- 77.** Which one is the hip joint? (1)
- A. Hinge joint
 - B. Ball and Socket joint
 - C. A and B
 - D. All

78. Muscles attached to bones by	(1)
A. Cartilage	B. Ligament
C. Tendon	D. Tissues
79. One of the following is not a part of the male reproductive system.	(1)
A. Penis	B. Testes
C. Uterus	D. Scrotum
80. Which of the following is not a part of the female reproductive system?	(1)
A. Ovary	B. Vagina
C. Sperm duct	D. All
81. What are the male sex cells?	(1)
A. Sperm cells	B. Scrotum
C. Sperm ducts	D. Vagina
82. The male sex cells made in	(1)
A. Testes	B. Penis
C. Sperm ducts	D. Ovary
83. The female sex cell is	(1)
A. Cilia	B. Cervix
C. Egg cells	D. Sperm cells
84. How long does the menstrual cycle take?	(1)
A. 5 days	B. 28 days
C. 14 days	D. 30 days
85. The character of an ovulation is	(1)
A. The release of a mature egg cell	
B. The loss of the lining of the uterus	
C. The joining of male sex cell and female sex cells	
D. Fertilization	
86. What is substance that passes through the placenta to the baby?	(1)
A. Oxygen	B. Carbon dioxide
C. Blood	D. Food
87. How long is gestation in humans?	(1)
A. 28 days	B. 9 years
C. 40 weeks	D. 11 Years
88. Which changes due to happen during puberty to boys and girls?	(1)
A. Hips widen	B. Underarm hair grows
C. Facial hair grows	D. Sperm production

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- 89.** STIs are most common in _ age groups. (1)
- A. Young and adults of ages 18-35 B. People of ages 60 and older
C. People of ages 36-45 D. All
-
- 90.** The overall number of cases of STIs is: (1)
- A. Rising B. Falling
C. Constant D. All
-
- 91.** As far as a person has no symptoms of STI, he/she (1)
- A. Cannot pass on STI B. Don't have STI
C. A and B D. None
-
- 92.** Which of the following is a cause for STIs in women? (1)
- A. Pelvic inflammatory disease B. Higher risk for cervical cancer
C. Ectopic pregnancy D. All of the above
-
- 93.** Which of the following sexually transmitted infection (STI) is caused by bacteria? (1)
- A. Gonorrhea B. Syphilis
C. A and B D. All
-
- 94.** One of the symptoms of genital herpes is (1)
- A. Burning in the genital area B. Painful blisters in the genital area
C. A and B D. None of the above
-
- 95.** Which one of the human organs untreated syphilis affects? (1)
- A. Heart B. Brain
C. Liver D. A and B
-
- 96.** Which of the followings is another type of STIs? (1)
- A. Chancroid B. Molluscum contagiosum
C. Scabies D. All of the above
-
- 97.** Which of the listed contraceptives requires prescriptions? (1)
- A. Birth control pill B. Contraceptive patch
C. Diaphragm D. All of the above
-
- 98.** It is one of the protective measures to lower the risk of getting an STI: (1)
- A. Use of male condoms
B. Eliminating multiple sex partners
C. Delay with having sexual relations as long as possible
D. All of the above
-
- 99.** What do male condoms offer that other forms of birth control do not? (1)
- A. Least chance of failure B. Best protection against STIs
C. Cheapest to use D. All

-
- 100.** Besides the condom, which one is another barrier method of birth control? (1)
- | | |
|---------------------|----------------------|
| A. Diaphragm | B. Withdrawal |
| C. IUD | D. All |
-
- 101.** Which one is a possible side effect of birth control pills? (1)
- | | |
|------------------------------|---------------------|
| A. Nausea | B. Headaches |
| C. Irregular bleeding | D. All |
-
- 102.** How long is the vaginal ring left in place? (1)
- | | |
|-------------------|-------------------|
| A. Months | B. 3 Weeks |
| C. 2 Weeks | D. 1 week |
-
- 103.** Which of the methods is a natural family planning? (1)
- A.** Basal temperature
 - B.** Tracking changes in cervical mucus
 - C.** Tracking the menstrual cycle on a calendar
 - D.** All
-
- 104.** How can you solve the problem of underage marriage of girls? (1)
- | | |
|------------------------------|---------------------|
| A. Awareness creation | B. Education |
| C. Law | D. All |
-
- 105.** Chewing Khat causes (1)
- A.** Hypertension, stomatitis, esophagitis, gastritis and constipation
 - B.** Acute coronary syndrome and myocardial infarction
 - C.** Metabolic disorders, Impotency, Diabetes mellitus, Hepatitis and Liver cirrhosis
 - D.** All of the above
-
- 106.** What types of drugs causing health hazards do you know? (1)
- | | |
|---------------------|-------------------|
| A. Marijuana | B. Cocaine |
| C. Heroin | D. All |
-
- 107.** While oxygen is carried by the red blood cells, most digested food and wastes are carried by the (1)
- | | |
|---------------------|-----------------------------|
| A. Plasma | B. White blood cells |
| C. Platelets | D. Red blood cells |
-
- 108.** What are the health hazards of using drugs? (1)
- A.** Abnormal heart attacks and increased liver damage or failure
 - B.** Weight loss and lung disease, stroke, confusion and brain damage
 - C.** Lung disease, seizures, stroke, mental confusion and brain damage
 - D.** All
-
- 109.** Which of the following is the correct pathway of breathing from the outer to the inner parts? (1)
- A.** Nasal cavity → trachea → bronchus → bronchioles → alveolus
 - B.** Alveolus → bronchus → bronchioles → trachea → nasal cavity
 - C.** Alveolus → bronchioles → bronchus → trachea → nasal cavity
 - D.** Nasal cavity → alveolus → bronchus → trachea → bronchioles

110. The absolute maximum amount of air you can take into or breathe out of your lungs is called (1)

- A.** Tidal volume **B.** Vital capacity
C. Residual volume **D.** Breathing hygiene

111. The right side chamber valve which directs the flow of blood by opening and closing is called (1)

- A.** Bicuspid valve **B.** Tricuspid valve
C. Mitral valve **D.** Vena cava

112. Which one of the following statements is true about pulmonary circulation? (1)

- A. Blood flows from the heart to all around the body and back again
- B. Carry only oxygenated blood from the body
- C. Blood flows from the heart to lungs and back again
- D. It transport oxygenated blood from the heart and back again

113. Which organic molecule in our body has functions as antigens & antibodies of immune system? (1)

- A. Lipids
B. Nucleic acids
C. Proteins
D. Carbohydrates

114. The blood group of human being which is commonly referred to as 'universal acceptor' is (1)

- A.** Blood group AB
- B.** Blood group O
- C.** Blood group A
- D.** Blood group B

115. How many bones does the textbook list as part of the total human skeleton (axial + appendicular)? (1)

- A. 180
B. 196
C. 206
D. 216

116. Which of the following best describes the axial skeleton? (1)

- A. The bones of the limbs and girdles
- B. The skull, rib cage, vertebral column and pelvis — forming the body's central core
- C. The bones of the arms and legs only
- D. The bones that support the teeth

117. Which type of muscle is described as involuntary and non-striated, often found in the walls of hollow organs such as the stomach and intestines? (1)

- A.** Skeletal muscle **B.** Cardiac muscle
- C.** Smooth muscle **D.** Myocardial muscle

118. Which muscle type is under conscious (voluntary) control and attaches to bones to enable movement? (1)

- A.** Cardiac muscle
- B.** Skeletal muscle
- C.** Smooth muscle
- D.** Involuntary striated muscle

119. Which structure is the site of spermatogenesis? (1)

- A. Fallopian tubes
B. Uterus
C. Seminiferous tubules
D. Prostate gland

120. Which of the following statements about cardiac muscle is correct? (1)

- A.** It is voluntary and striated.
- B.** It is involuntary and striated.
- C.** It is voluntary and non-striated.
- D.** It is involuntary and non-striated.

-
- 121.** What is the main difference between skeletal muscle and smooth muscle? (1)
- A. Skeletal muscle is non-striated and involuntary; smooth muscle is striated and voluntary.
 - B. Skeletal muscle is striated and voluntary; smooth muscle is non-striated and involuntary.
 - C. Both are striated but only skeletal muscle is voluntary.
 - D. Both are non-striated but only smooth muscle is voluntary.
-
- 122.** Which of the following bones belong to the axial skeleton? (1)
- A. Femur and humerus
 - B. Skull and sternum
 - C. Clavicle and scapula
 - D. Pelvis only
-
- 123.** Which of these statements about the human skeletal system is incorrect? (1)
- A. The axial skeleton includes the skull, vertebral column, rib cage, and sternum.
 - B. The appendicular skeleton consists of the limbs and their girdles.
 - C. All muscles in the human body are voluntary.
 - D. The thoracic cage helps protect vital organs such as the heart and lungs.
-
- 124.** Which of the following is considered a harmful traditional practice because it can cause severe physical and psychological damage? (1)
- A. Ear piercing
 - B. Female genital mutilation (FGM)
 - C. Hair braiding
 - D. Traditional greetings
-
- 125.** The primary function of the thoracic (rib) cage is to: (1)
- A. Support body posture
 - B. Protect the heart and lungs
 - C. Protect the brain
 - D. Aid in digestion
-
- 126.** Which contraceptive method involves preventing ovulation through hormonal regulation? (1)
- A. Male condom
 - B. Oral contraceptive pills
 - C. Intrauterine device (IUD)
 - D. Withdrawal method
-
- 127.** The muscle that separate the chest cavity from the abdominal cavity is known as (1)
- A. Nasal cavity
 - B. Trachoma
 - C. Diaphragm
 - D. Epiglottis
-
- 128.** Which of the following correctly describes gametogenesis? (1)
- A. The fusion of sperm and ovum to form a zygote
 - B. The process of forming haploid gametes through meiosis
 - C. The growth of the embryo after fertilization
 - D. The process where diploid gametes are produced by mitosis
-
- 129.** When you breathe out during gas exchange, all of the following happens to the breathing movement except? (1)
- A. The volume of the thorax decrease
 - B. Diaphragm moves upward
 - C. Pressure inside chest cavity decreases
 - D. Pressure inside chest cavity increases
-
- 130.** Which of the following is the correct pathway of breathing from the outer to the inner parts? (1)
- A. Nasal cavity→trachea→bronchus→bronchioles→alveolus
 - B. Alveolus→bronchus→bronchioles→trachea→nasal cavity
 - C. Alveolus→bronchioles→bronchus→trachea→nasal cavity
 - D. Nasal cavity→alveolus→bronchus→trachea→bronchioles

131.	Which of the following is the types of blood vessels that carries deoxygenated blood from the heart to lungs is called	(1)
A.	Systematic circulation	B. Pulmonary vein
C.	Pulmonary circulation	D. Pulmonary artery
132.	The right side chamber valve which directs the flow of blood by opening and closing is called	(1)
A.	Bicuspid valve	B. Tricuspid valve
C.	Mitral valve	D. Vena cava
133.	In grade 9 laboratory class, the teacher gives the students a sampler of food and asked them to investigate the presence of glucose. What kind of test and result will enable the students to confirm the presence of glucose respectively?	(1)
A.	Benedict's test and orange colour	B. Biuret test and purple colour
C.	Emulsion test and milky white colour	D. Iodine test and blue black colour
134.	In a Benedict's test for sucrose, why does the test sample first boil with hydrochloric acid?	(1)
A.	To reduce Benedict's solution	B. To neutralize the solution
C.	To hydrolyze non-reducing sugar	D. To retest mixture with Benedict solution
135.	Which deficient nutrients are correctly paired with their deficiency diseases?	(1)
A.	Protein→Scurvy	B. Carbohydrate→Beri-beri
C.	Iron→Marasmus	D. Vitamin A→Night blindness
136.	The muscle that separate the chest cavity from the abdominal cavity is known as	(1)
A.	Nasal cavity	B. Trachoma
C.	Diaphragm	D. Epiglottis
137.	When you breathe out during gas exchange, all of the following happens to the breathing movement except?	(1)
A.	The volume of the thorax decrease	B. Pressure inside chest cavity decreases
C.	Diaphragm moves upward	D. Pressure inside chest cavity increases
138.	Which chamber of the heart receives oxygenated blood from the lungs?	(1)
A.	Left atrium	B. Right atrium
C.	Left ventricle	D. Right ventricle
139.	The exchange of gases (oxygen and carbon dioxide) occurs in which part of the respiratory system?	(1)
A.	Bronchi	B. Lungs
C.	Trachea	D. Nasal cavity
140.	Which of the following products are formed during anaerobic respiration of one glucose molecule in animal cells?	(1)
A.	Ethanol, NAD, 2ATP	B. Lactic acid, CO ₂ , 2ATP
C.	Alcohol, ADP, 2ATP	D. Lactic acid, O ₂ , CO ₂
141.	Which blood vessel carries oxygenated blood from the heart to the body?	(1)
A.	Veins	B. Arteries
C.	Capillaries	D. Lymphatic vessels

-
- 142.** Which of the following structures prevents food from entering the trachea? (1)
- | | |
|------------|---------------|
| A. Larynx | B. Epiglottis |
| C. Pharynx | D. Alveoli |
-
- 143.** What is the main function of the digestive system? (1)
- | | |
|---------------------|--|
| A. To absorb oxygen | B. To break down food and absorb nutrients |
| C. To pump blood | D. To regulate body temperature |
-
- 144.** Which part of the digestive system is primarily responsible for the absorption of nutrients? (1)
- | | |
|--------------------|--------------------|
| A. Stomach | B. Large intestine |
| C. Small intestine | D. Liver |
-
- 145.** Which enzyme in the stomach helps break down proteins? (1)
- | | |
|------------|------------|
| A. Amylase | B. Lipase |
| C. Pepsin | D. Trypsin |
-
- 146.** Where does the digestion of carbohydrates begin? (1)
- | | |
|--------------------|--------------------|
| A. Stomach | B. Mouth |
| C. Small intestine | D. Large intestine |
-
- 147.** Which organ is responsible for producing bile? (1)
- | | |
|-------------|--------------------|
| A. Stomach | B. Liver |
| C. Pancreas | D. Small intestine |
-
- 148.** Which of the following structures is involved in the filtration of blood in the kidneys? (1)
- | | |
|---------------|-------------|
| A. Nephron | B. Alveolus |
| C. Glomerulus | D. Glucose |
-
- 149.** Which of the following is a function of the large intestine? (1)
- | | |
|---------------------------------|---|
| A. Absorption of most nutrients | B. Absorption of water and electrolytes |
| C. Storage of bile | D. Digestion of proteins |
-
- 150.** The following all are uses of mitosis division except? (1)
- | | |
|---------------------------|------------------------------|
| A. Replace worn out cells | B. Increase number of cells |
| C. Repair damaged cells | D. Produce reproduction cell |
-
- 151.** The process of formation of ova by meiosis process is known as (1)
- | | |
|--------------------|---------------------|
| A. Spermatogenesis | B. Oogenesis |
| C. Cell fraction | D. Somatic division |
-
- 152.** The mitosis stage in which the sister chromatids are aligned in equator are known by (1)
- | | |
|--------------|--------------|
| A. Anaphase | B. Metaphase |
| C. Telophase | D. Prophase |

-
- 153.** When the pure round yellow seed pea plant is crossed by di-hybrid cross with pure wrinkled, the phenotype ratio of filial generation or f1 generation based on round yellow round green wrinkled yellow and wrinkled green respectively is (1)
- A. 3:3:9:1
B. 9:3:3:1
C. 3:9:3:1
D. 1:3:3:9
-
- 154.** The tabular structural method that show the distribution of different traits throughout generation or offspring is known as (1)
- A. Pedigree tree
B. Punnet square
C. Phylogenetic tree
D. None of the above
-
- 155.** The type of dominance in genetics that shows intermediate characters between or among different offspring and parental type is (1)
- A. Concomitance
B. Complete dominance
C. Incomplete dominance
D. All of the above
-
- 156.** The process that leads up to the cell dividing into two during mitosis is known as (1)
- A. Prophase
B. Interphase
C. Cytokinesis
D. Metaphase
-
- 157.** Which one of the following is not characteristics of the daughter cells produced by meiosis (1)
- A. There are four daughter cells produced
B. They are genetically the same or without difference
C. They are haploid
D. May be gametes because of gametogenesis
-
- 158.** Among the following, one is not differentiated cell. (1)
- A. Blood cells produced by mitosis
B. Nerve cell produced by mitosis
C. Embryonic stem cells produced by mitosis
D. Muscle cells produce by mitosis
-
- 159.** One of the following is different from other (1)
- A. Skin color
B. Hair color
C. Eye pigment
D. Absence of melanin in our skin (albinism)
-
- 160.** How many chromosomes are in human sperm and eggs? (1)
- A. 30
B. 23
C. 46
D. 92
-
- 161.** When meiosis is stopped in male and females human being respectively (1)
- A. At death and menopause
B. At death for both
C. At menopause stage and at death
D. At puberty time in both sex
-
- 162.** The position of each gene within their long chromosomal position is known as ? (1)
- A. Loci
B. Homozygous
C. Heterozygosity
D. Phenotype

163. The rough and physical appearance of traits by their allelomorphism expression is known as ?	(1)
A. Genotype	B. Phenotype
C. Homologous type	D. Heterozygous type
164. Which process results in the formation of gametes?	(1)
A. Mitosis	B. Meiosis
C. Fertilization	D. DNA replication
165. During which phase of mitosis do sister chromatids separate?	(1)
A. Prophase	B. Metaphase
C. Anaphase	D. Telophase
166. Gregor Mendel is known for his work on:	(1)
A. Evolution	B. Inheritance of traits in pea plants
C. Natural selection	D. Chromosome mapping
167. An organism with two identical alleles for a trait is:	(1)
A. Heterozygous	B. Homozygous
C. Dominant	D. Recessive
168. The physical expression of a gene is known as:	(1)
A. Genotype	B. Phenotype
C. Allele	D. Trait
169. In humans, the sex chromosomes are:	(1)
A. XX in males and XY in females	B. XY in males and XX in females
C. YY in males and XX in females	D. XY in both males and females
170. The structure that carries genetic information is:	(1)
A. Ribosome	B. Mitochondrion
C. Chromosome	D. Lysosome
171. Which of the following best describes a gene?	(1)
A. A type of protein	
B. A segment of DNA that codes for a protein	
C. A carbohydrate molecule	
D. A lipid molecule	
172. What is the probability of producing a homozygous recessive offspring from two heterozygous parents?	(1)
A. 25%	B. 50%
C. 75%	D. 100%
173. Which term describes different forms of a gene?	(1)
A. Loci	B. Alleles
C. Chromatids	D. Genomes

-
- 174.** The law stating that allele pairs separate during gamete formation is: (1)
- A. Law of Independent Assortment B. Law of Segregation
C. Law of Dominance D. Law of Recessiveness
-
- 175.** A Punnett square is used to: (1)
- A. Determine DNA sequences B. Predict the outcome of genetic crosses
C. Measure cell division D. Identify mutations
-
- 176.** Which of the following is true about dominant alleles? (1)
- A. They are only expressed in the homozygous state
B. They are never expressed
C. They mask the expression of recessive alleles
D. They are less common in populations
-
- 177.** In a dihybrid cross, how many different phenotypes can be produced? (1)
- A. 2 B. 4
C. 8 D. 16
-
- 178.** Which process increases genetic variation in offspring? (1)
- A. Mitosis B. Asexual reproduction
C. Meiosis D. Cloning
-
- 179.** The observable characteristics of an organism are influenced by: (1)
- A. Only its genotype B. Only its environment
C. Both genotype and environment D. Neither genotype nor environment
-
- 180.** A carrier of a recessive genetic disorder: (1)
- A. Shows symptoms of the disorder B. Cannot pass the disorder to offspring
C. Has one copy of the mutated gene D. Has two copies of the mutated gene
-
- 181.** Selective breeding involves: (1)
- A. Random mating B. Choosing parents with desired traits
C. Genetic engineering D. Natural selection

— END OF EXAMINATION —